Second-order entropy stability of a three-mode fermionic product channel

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Abstract

The minimum output entropy of a fermionic product channel is known over generalized Gaussian inputs, but this restricted optimization does not control arbitrary non-Gaussian perturbations. We study the complete three-mode channel on the eight-dimensional qubit Hilbert space, including odd Majorana monomials and parity-coherent inputs. For every ordered attenuation vector $0 \le b_6 \le \cdots \le b_1 < 1$, we prove that the known generalized Gaussian optimizer $|+++\rangle$ is stationary in all complex pure-state directions and has a positive-semidefinite real entropy Hessian. Its twelve-dimensional Gaussian tangent decouples from the two-dimensional non-Gaussian normal space. In the normalized chart with normal vector $|-+-\rangle$, the two normal curvatures obey $H_{\mathrm{Im}} \ge H_{\mathrm{Re}} \ge 2b_1^2(1-b_1^2)^4/(65\ln 2)$, and are strictly positive when $b_1 > 0$. The proof combines a logarithmic resolvent reduction with a complete exact rational Bernstein certificate for four polynomials on an ordered parameter chamber. The certificate and two algebraically distinct replay implementations are supplied. Zero and repeated smaller attenuations are included. This result excludes a quadratic non-Gaussian entropy descent mechanism at the Gaussian encoder; it does not establish a global minimum, higher-mode stability, or nonlinear local minimality along Gaussian zero modes. The rank-singular face $b_1 = 1$ is outside the differentiability domain considered here.

Keywords: quantum channel; minimum output entropy; Majorana attenuation; generalized Gaussian state; exact polynomial certificate

1 Introduction

Fermionic product channels provide a finite-dimensional family in which noise acts multiplicatively on a complete Majorana operator basis. Bravyi introduced these channels, determined their minimum output entropy over generalized Gaussian inputs, and proved unrestricted optimality for at most two modes [1]. For three and four modes, numerical iterations supported the same global formula. The Gaussian optimization and finite numerical evidence leave a distinct local question: can a non-Gaussian perturbation lower the output entropy to quadratic order at the proposed encoder?

We answer this local question for every ordered attenuation vector in the three-mode chamber with a full-rank reference output. At the encoder $g = |+++\rangle$, there are fourteen real projective pure-state directions. Twelve are tangent to the generalized Gaussian orbit; the two remaining directions are represented by $\chi = |-+-\rangle$ and $\mathrm{i}\chi$. The full entropy gradient vanishes, these two sectors have no mixed Hessian entries, and both normal curvatures have an explicit positive lower bound whenever the channel is not completely depolarizing. The known Gaussian optimum then supplies the nonnegative tangent block.

The central step is a model-specific sign certificate. We reduce the real normal curvature to an integral whose numerator is cubic in a resolvent variable. Each of its four coefficients has

a complete rational Bernstein expansion with nonnegative entries on a five-dimensional ratio cube. A positive coefficient slice gives strictness and a conservative quantitative bound. The calculation uses exact integers and rationals, without fitting, sampling, or a noise-parameter expansion.

Several nearby results concern different quantities. Completely bounded output entropy and entanglement-assisted capacity for radial fermion multipliers [2] do not identify this ordinary output-entropy Hessian. The Fourier-multiplier transference results in [3] also require their stated algebraic hypotheses. Parity-constrained fermionic channel optimizations and composition in [4] use a different input domain. Standard local-additivity methods [5] provide useful entropy derivatives but do not supply an optimizer for this channel. Our contribution is the continuous-parameter three-mode second variation and its exact certificate.

2 Channel, input domain, and statement

2.1 The complete attenuation map

On $\mathcal{H} = (\mathbb{C}^2)^{\otimes 3}$, with the first qubit most significant in the computational basis, define

$$c_{2j-1} = Z_1 \cdots Z_{j-1} X_j, \qquad c_{2j} = Z_1 \cdots Z_{j-1} Y_j, \qquad j = 1, 2, 3. \tag{1}$$

These Hermitian Majoranas satisfy $\{c_p, c_q\} = 2\delta_{pq} I$. For $x \in \{0,1\}^6$, let $c(x) = c_1^{x_1} \cdots c_6^{x_6}$, with factors in increasing order. The channel is defined on all of M_8 by

$$\Phi_{\boldsymbol{b}}(c(x)) = \left(\prod_{p : x_p = 1} b_p \right) c(x), \qquad \Phi_{\boldsymbol{b}}(I) = I. \tag{2}$$

In particular, the definition retains odd and higher-order monomials; it is not merely an action on a covariance matrix.

For $0 \le b_p \le 1$, complete positivity and trace preservation follow from Bravyi's factor representation [1, Eqs. (9)–(10)]:

$$P = (-\mathrm{i})^3 c_1 \cdots c_6 = Z_1 Z_2 Z_3, \quad K_p = \mathrm{i} P c_p, \quad \Phi_p(\rho) = \frac{(1+b_p)\rho + (1-b_p) K_p \rho K_p}{2}. \tag{3}$$

Each K_p is a Hermitian unitary; conjugation by K_p changes only the sign of c_p. Thus the commuting factor superoperators compose to (2). The name "product" refers to these factors, which are not independent channels on the three physical qubits.

The optimization admits every qubit density matrix, including complex pure states with parity coherence. No definite-parity restriction is imposed. This is the encoded spin/qubit problem of [1]; it does not assert an operational preparation or communication rate for isolated fermions subject to parity superselection. Attenuations and state coordinates are dimensionless. We use $S(\rho) = -\operatorname{Tr} \rho \log_2 \rho$, so entropy and its coordinate curvature are in bits. There is no Hamiltonian, boundary condition, thermal scale, continuum limit, or perturbative dynamical approximation in this model.

2.2 Gaussian optimum and the pure-state chart

"Generalized Gaussian" follows Definition 1 of [1]: the allowed unitary generators include Hermitian linear and quadratic Majorana combinations, rather than quadratic terms alone. Write

$$(a, b, c, d, e, f) = (b_1, b_2, b_3, b_4, b_5, b_6), \qquad 0 \le f \le e \le d \le c \le b \le a < 1. \tag{4}$$

The pure input

$$|g\rangle = |+++\rangle, \qquad |\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}, \tag{5}$$

is the state in Bravyi's Eq. (36). Indeed its three commuting stabilizers can be written $c_1 = X_1$, $-\mathrm{i}c_2c_3 = X_1X_2$, and $-\mathrm{i}c_4c_5 = X_2X_3$. The source Gaussian optimization gives

$$S(\Phi_{\boldsymbol{b}}(|g\rangle\langle g|)) = h_2\left(\frac{1+a}{2}\right) + h_2\left(\frac{1+bc}{2}\right) + h_2\left(\frac{1+de}{2}\right), \tag{6}$$

where $h_2(u) = -u\log_2 u - (1-u)\log_2(1-u)$. This is the known minimum over generalized Gaussian inputs. Equality with the unrestricted global minimum is not assumed.

Let $|s(q)\rangle$, $q = 0, \ldots, 7$, be the orthonormal X basis, where the binary digits of q specify minus signs and $|s(0)\rangle = g$. A projective chart at g is

$$|\psi(w)\rangle = \frac{|g\rangle + \sum_{q=1}^{7} w_q|s(q)\rangle}{\sqrt{1+\sum_{q=1}^{7}|w_q|^2}}, \qquad w_q = u_q + \mathrm{i}v_q. \tag{7}$$

The Hessian below uses the fourteen real coordinates u_q, v_q; their tangent vectors have unit Hilbert norm. Define $\mathcal{F}(w) = S(\Phi_{\boldsymbol{b}}(|\psi(w)\rangle\langle\psi(w)|))$. Its Gaussian tangent is understood as a real subspace of this projective tangent, with inner product $\mathrm{Re}\langle v, w\rangle$.

Theorem 1 (Three-mode second variation)**.** *For every attenuation vector in* (4)*, $\nabla\mathcal{F}(0) = 0$ and $\nabla^2\mathcal{F}(0)$ is positive semidefinite. The twelve-dimensional generalized Gaussian tangent and its normal space*

$$\mathcal{N} = \mathrm{span}_{\mathbb{R}}\{|\chi\rangle, \mathrm{i}|\chi\rangle\}, \qquad |\chi\rangle = |-++-\rangle, \tag{8}$$

have zero mixed Hessian block. In the normal chart $\psi(z) = (g + z\chi)/\sqrt{1+|z|^2}$, the real and imaginary coordinate directions diagonalize that block and satisfy

$$H_{\mathrm{Im}} \geq H_{\mathrm{Re}} \geq \frac{2a^2(1-a^2)^4}{65\ln 2}. \tag{9}$$

For $a > 0$ both normal curvatures are strictly positive. At $a = 0$ the complete channel is depolarizing and the full Hessian is zero.

Here $H_{\mathrm{Re}} = \partial_x^2\mathcal{F}(z)|_{z=0}$ for $z = x + \mathrm{i}y$, and similarly for H_{Im}. Repeated and zero smaller attenuations belong to the theorem. The face $a = 1$ is excluded because the reference output can be rank deficient. Positive-semidefinite curvature does not imply nonlinear local minimality along zero-curvature Gaussian directions.

3 Stationarity and the normal Hessian

3.1 Output spectrum and coherence

For an X-basis sign triple $s = (s_1, s_2, s_3)$, introduce the bijection

$$t = (s_1, s_1s_2, s_2s_3), \qquad r = (a, bc, de). \tag{10}$$

The reference output is diagonal in this basis, with eigenvalues

$$\lambda(t) = \frac{1}{8}(1+at_1)(1+bct_2)(1+det_3). \tag{11}$$

All are strictly positive under (4), even when smaller attenuations vanish or coincide. The normal input output $\Phi_{\boldsymbol{b}}(|\chi\rangle\langle\chi|)$ is diagonal with eigenvalues $\lambda(-t)$. The corresponding index pairing flips the first and third X signs, and therefore changes t to $-t$.

For $E = |\chi\rangle\langle g|$, the only nonzero entries of $\Phi_{\boldsymbol{b}}(E)$ join those paired basis vectors. With $s(t)$ the inverse sign map,

$$A(t) := \langle s(-t)|\Phi_{\boldsymbol{b}}(E)|s(t)\rangle = \frac{t_3 f}{8}(a+t_1)(b+t_2c)(d+t_3e). \tag{12}$$

Equations (11)–(12) follow by expanding the three input matrices in the complete Majorana basis, or by applying the six CP factors (3). Both constructions are reproduced exactly in Online Resource 1, including all 192 matrix entries rather than only the covariance data.

3.2 Selection rules and tangent splitting

Every ordered Majorana monomial is a Pauli string up to a scalar phase. Hence (2) is diagonal on the Pauli basis. A Pauli string in the X basis has a fixed binary flip pattern: its nonzero matrix entries satisfy $p \oplus q = h$. For each h, the Pauli strings of that pattern span the matrices with that support. The channel therefore preserves each flip pattern.

The first density variation in any direction $|s(q)\rangle$, $q \neq 0$, has pattern q and no output diagonal. Since the reference log-output is diagonal, every first entropy derivative vanishes. For distinct patterns, the first variations are orthogonal in the logarithmic derivative bilinear form. Their mixed second density variation has the nonzero pattern $p \oplus q$ and thus zero pairing with the diagonal log-output. The Hessian has no entries mixing distinct flip patterns.

The projective vectors obtained by acting with $\mathrm{i}c_p$ and $\mathrm{i}(\mathrm{i}c_p c_q)$ on g, deleting the component along g, give both real directions for the six patterns

$$\{001, 010, 011, 100, 110, 111\}. \tag{13}$$

They have real rank twelve and no component with pattern 101. This is a finite calculation with the six linear and fifteen quadratic generators in (1); the supplement checks the rank exactly. These generators close to the generalized Gaussian Lie algebra. Thus (13) is its tangent and 101, namely χ, is its complex orthogonal complement. The flip selection rule proves the Gaussian–normal mixed block is zero for every attenuation vector.

3.3 Entropy derivative in the normalized chart

For $x, y > 0$ define the logarithmic divided difference

$$L(x,y) = \begin{cases} (\ln x - \ln y)/(x-y), & x \neq y, \\ 1/x, & x = y, \end{cases} \qquad L(x,y) = \int_0^\infty \frac{\mathrm{d}v}{(v+x)(v+y)}. \tag{14}$$

The continuous value is essential when output eigenvalues coincide. For a trace-preserving twice-differentiable density path $\sigma(\epsilon)$ through a positive diagonal σ_0 with eigenvalues λ_i,

$$\left.\frac{\mathrm{d}^2 S}{\mathrm{d}\epsilon^2}\right|_0 = -\frac{1}{\ln 2}\left[\operatorname{Tr}\sigma''(0)\ln\sigma_0 + \sum_{i,j} |\sigma'(0)_{ij}|^2 L(\lambda_i,\lambda_j)\right]. \tag{15}$$

This standard resolvent derivative is also used in local entropy analysis [5, Sec. 3].

For either real normal coordinate, the second density variation is

$$\sigma''(0) = 2\big[\Phi_{\boldsymbol{b}}(|\chi\rangle\langle\chi|) - \Phi_{\boldsymbol{b}}(|g\rangle\langle g|)\big]. \tag{16}$$

It must be retained: differentiating the coherence alone would give the wrong sign problem. Combining (15) with the paired diagonal entries gives

$$H_{\mathrm{Re}} = \frac{1}{\ln 2}\sum_t \Big\{[\lambda(t)-\lambda(-t)]^2 - [A(t)+A(-t)]^2\Big\} L(\lambda(t),\lambda(-t)), \tag{17}$$

$$H_{\mathrm{Im}} = \frac{1}{\ln 2}\sum_t \Big\{[\lambda(t)-\lambda(-t)]^2 - [A(t)-A(-t)]^2\Big\} L(\lambda(t),\lambda(-t)). \tag{18}$$

The sums run over all eight sign triples and count both members of each pair. Real and imaginary first variations have real and purely imaginary matrix entries, respectively, so their mixed entry vanishes. Moreover,

$$A(t)A(-t) = \frac{f^2(1-a^2)(b^2-c^2)(d^2-e^2)}{64} \geq 0. \tag{19}$$

It follows directly that $H_{\mathrm{Im}} \geq H_{\mathrm{Re}}$. Repeated pairs may make the two curvatures equal; their positivity will follow without division by a parameter difference.

4 Exact positivity over the ordered chamber

4.1 Four resolvent coefficients

Set $r = bc$ and $s = de$ in this section. Represent the four unordered pairs by $t_1 = 1$, $t_2, t_3 \in \{1, -1\}$, indexed by i. Define

$$W_i = 1 + art_2 + ast_3 + rst_2t_3, \tag{20}$$

$$D_i = a + rt_2 + st_3 + arst_2t_3, \tag{21}$$

$$V_i = bd + cet_2t_3 + a(cdt_2 + bet_3). \tag{22}$$

For the corresponding pair, $\lambda + \lambda' = W_i/4$, $\lambda - \lambda' = D_i/4$, and $A + A' = t_3 f V_i/4$. In particular $W_i > 0$ and

$$G = \lambda(t)\lambda(-t) = \frac{(1-a^2)(1-r^2)(1-s^2)}{64} > 0 \tag{23}$$

is independent of the pair. For $v \geq 0$, put

$$q = v^2 + G, \qquad z = \frac{v}{4q}. \tag{24}$$

Using (14) in (17) and combining the four fractions yields

$$H_{\mathrm{Re}} = \frac{1}{8\ln 2}\int_0^\infty \frac{N(z)}{q\prod_{i=1}^4(1+zW_i)}\,\mathrm{d}v, \qquad N(z) = \sum_{i=1}^4 (D_i^2 - f^2V_i^2)\prod_{j\neq i}(1+zW_j). \tag{25}$$

The numerator has degree at most three in z. The coefficient N_k of z^k has derivative in f^2 equal to

$$\frac{\partial N_k}{\partial(f^2)} = -\sum_i V_i^2 e_k(W_1, \ldots, \widehat{W_i}, \ldots, W_4) \leq 0, \tag{26}$$

where e_k is the elementary symmetric polynomial and $e_0 = 1$. All its arguments are positive. Since $f \leq e$, it suffices to certify every coefficient at $f = e$.

4.2 The complete rational certificate

The sign summation in (25) makes every attenuation exponent even, as can be verified by direct expansion. At $f = e$, put $A = a^2$, $B = b^2$, $C = c^2$, $D = d^2$, and $E = e^2$. Map the ordered chamber to a cube by

$$A = x_0, \quad B = x_0x_1, \quad C = x_0x_1x_2, \quad D = x_0x_1x_2x_3, \quad E = x_0x_1x_2x_3x_4, \quad 0 \leq x_j \leq 1. \tag{27}$$

This covers the zero faces: if a preceding parameter is zero, all following parameters are zero and their remaining ratios may be chosen as zero. No division by a vanishing attenuation is required.

Let $P_k(x)$ denote N_k after this substitution. With $B_i^n(x) = \binom{n}{i}x^i(1-x)^{n-i}$, its tensor expansion is

$$P_k(x) = \sum_{\boldsymbol{i}} \beta_{\boldsymbol{i}}^{(k)} \prod_{j=0}^4 B_{i_j}^{n_j^{(k)}}(x_j). \tag{28}$$

If $p_{\boldsymbol{m}}$ are the exact power coefficients, the values are computed by the finite identity

$$\beta_{\boldsymbol{i}}^{(k)} = \sum_{\boldsymbol{m}\leq\boldsymbol{i}} p_{\boldsymbol{m}} \prod_{j=0}^4 \frac{\binom{i_j}{m_j}}{\binom{n_j^{(k)}}{m_j}}. \tag{29}$$

Table 1: Complete exact tensor Bernstein witness for the four coefficients of the real-normal resolvent numerator at $f = e$. Entries are stored as integer numerators divided by the indicated common denominator. All 37,320 entries are nonnegative; degrees are derived from expanded polynomials, rather than inferred from samples.

k	$(n_0, n_1, n_2, n_3, n_4)$	Entries	Denominator	Zeros
0	$(5, 4, 3, 2, 2)$	1,080	1,440	181
1	$(6, 5, 4, 3, 2)$	2,520	43,200	366
2	$(9, 8, 6, 4, 3)$	12,600	152,409,600	1,295
3	$(10, 9, 7, 5, 3)$	21,120	2,000,376,000	2,066

Table 1 gives the derived finite degrees, complete coefficient counts, common positive denominators, and numbers of zero entries. Every other entry is strictly positive.

Online Resource 1 supplies the entire witness, the exact polynomial generator, the forward basis conversion, and two replay implementations. One rebuilds the full Majorana monomial channel and uses separate sparse integer polynomial convolution followed by inverse finite differences. On each axis the inverse identity is $p_j = \binom{n}{j}\Delta^j\beta_0$; applying it successively verifies every multivariate power coefficient. The other builds the CP-factor channel directly, extracts the four summands from its matrix entries, and recomputes every Bernstein value using rational arithmetic before comparison. The matrix entries, real tangent rank, flip-support selection rules, and known one-/two-mode spectra are checked as well. No polynomial cache is used by either replay to establish the witness identity. The supplied certificate is an exact finite algebraic witness, not a numerical scan or a fitted reconstruction.

4.3 Strictness and the quantitative bound

For $k = 0$, every coefficient with $i_0 = 1$ in (28) equals $4/5$, independent of the other four indices. Nonnegativity of all other entries and partition of unity in the remaining axes give

$$P_0(x) \geq \frac{4}{5} B_1^5(A) = 4A(1 - A)^4. \tag{30}$$

The other P_k are nonnegative on the cube. By (26), the original $f \leq e$ numerator satisfies

$$N(z) \geq 4a^2(1 - a^2)^4 \qquad (z \geq 0). \tag{31}$$

The positive denominator in (25) proves strict real-normal positivity when $0 < a < 1$.

To obtain an explicit constant, restrict the integral to $v \geq 1$. From $G \leq 1/64$, $W_i \leq 4$, and $z \leq 1/(4v)$,

$$q \prod_i (1 + zW_i) \leq \frac{65}{64} v^2 (1 + 1/v)^4 \leq \frac{65}{4} v^2. \tag{32}$$

Substitution in (25) and $\int_1^\infty v^{-2} \mathrm{d}v = 1$ yield $H_{\mathrm{Re}} \geq 2a^2(1 - a^2)^4/(65 \ln 2)$. Together with (19), this proves (9). The bound is deliberately conservative; it is uniform in all five smaller attenuations, including their degeneracies.

5 Completion of the full Hessian and interpretation

Bravyi's Gaussian minimum [1, Theorem 1 and Eq. (34)] makes the restricted Hessian on the Gaussian orbit nonnegative. For a curved path on that orbit, its acceleration would contribute the full gradient paired with the path's second tangent derivative. The full gradient is zero, so the restricted Hessian equals the restriction of the projective Hessian (7). The twelve tangent

directions and the two normal directions exhaust the real projective tangent. Their mixed block vanishes, and the normal block is nonnegative by the certificate. This proves the full positive-semidefinite statement in Theorem 1. If $a = 0$, ordering forces every attenuation to be zero; the channel sends all density matrices to $I/8$ and every derivative vanishes.

The calculation controls a concrete failure mechanism for the source global conjecture. A negative non-Gaussian Hessian direction at g would immediately lower its output entropy to quadratic order. Theorem 1 excludes that route throughout (4). It also supplies an exact diagnostic for an entropy optimizer: a negative normal curvature at this encoder in the stated chart signals an implementation or convention error, rather than a counterexample. The normalization term (16) and the repeated-eigenvalue limit in (14) are particularly relevant to that diagnostic.

For the unrestricted covariant qubit channel, Bravyi's known single-use Holevo relation is $C_1 = 3 - S_{\min}$ [1]. The local obstruction constrains quadratic improvements of the Gaussian encoder in that optimization, but establishes no new value of $S_{\min}$ or C_1. It also gives no regularized classical, quantum, private, entanglement-assisted, or parity-constrained rate.

The remaining limitations are substantial. A positive-semidefinite Hessian may have flat Gaussian directions, for which higher input-state orders can matter. We have neither controlled those orders nor excluded distant non-Gaussian minima. The theorem is for three modes, not every finite mode number, and does not extend entropy differentiability to the singular face $a = 1$. The unrestricted all-mode global formula is therefore left unresolved by this work. No priority is claimed for generalized Gaussian structure, the logarithmic resolvent, or the Bernstein basis; the result is their exact model-specific application to the two transverse entropy curvatures.

Statements and declarations

Computational material. Online Resource 1 is the accompanying source supplement containing `certificate.json`, its generating programs, exact replay programs, pinned environment requirements, and reproduction instructions. The complete sign argument uses exact integer and rational arithmetic; the included floating Hessian checks are supplementary diagnostics. No external dataset is required. The original channel is specified by (1)–(3) and attributed to [1].

Artificial intelligence disclosure. OpenAI Codex was used to propose and develop the analytic reduction, generate the computational code, execute exact checks, screen literature, and draft the manuscript. A separate Codex session performed an internal scientific audit of the result. These uses extend beyond copy editing. The internal audit is neither external peer review nor human verification. Human scientific verification, authorship confirmation, and assessment of the publisher's AI policy remain pending for this local draft.

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